
When Are Teacher Tokens Reliable?

Position-Weighted On-Policy Self-Distillation for Reasoning

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Abstract

On-policy self-distillation (OPSD) trains a student on its own rollouts using a privileged teacher, but its standard objective weights all generated tokens equally, implicitly treating the privileged teacher target as equally reliable at every student-visited prefix. Existing entropy-based OPD methods relax this uniformity by modulating token-level supervision with teacher entropy, but high teacher entropy in reasoning has an ambiguous reliability meaning: it can reflect either non-viable uncertainty or benign solution diversity. To identify this phenomenon, we introduce a branch-viability diagnostic. Specifically, we record next-token alternatives from the privileged-answer teacher prompt, force each alternative after the student prompt plus its on-policy spine prefix, and test whether the resulting student-template continuation recovers the correct answer. On Qwen3-4B, we find that an oriented within-sequence position score is the strongest tested predictor of teacher-token reliability, reaching an area-under-ROC-curve (AUROC) of 0.83 with a 95% cluster-bootstrap interval of [0.66, 0.95]; local uncertainty scores are at most 0.57. Motivated by this trajectory-level structure, we propose Position-Weighted On-Policy Self-Distillation (PW-OPSD), which applies an increasing position weight while keeping the same student rollout, privileged teacher pass, and clipped forward-KL target as OPSD. In our comprehensive evaluations with different random seeds, the diagnostic-derived PW-OPSD improves AIME 2024 and AIME 2025 Avg@12 by +1.0 and +1.1 points, and a generalization evaluation on two larger-scale models from different families, DeepSeek-R1-Distill-Llama-8B and Olmo-3-7B-Think, also demonstrates consistent aggregate Avg@12 improvements. These results show that teacher-token reliability in reasoning distillation is trajectory-structured and can be utilized without additional teacher computation. The code is available at <https://github.com/SaFo-Lab/PW-OPSD>

1 Introduction

On-policy self-distillation (OPSD) [Zhao et al., 2026] is a practical recipe for distilling mathematical reasoning because it supervises the student on prefixes the student actually visits. The student samples a rollout from the ordinary problem prompt, and a privileged copy of the model, conditioned on reference information, provides token-level targets along that same rollout. This construction avoids the trajectory mismatch of teacher-generated demonstrations while still injecting privileged information at training time.

The standard OPSD objective nevertheless makes an implicit reliability assumption: every generated token receives the same forward-KL supervision [Zhao et al., 2026]. This uniform objective treats the

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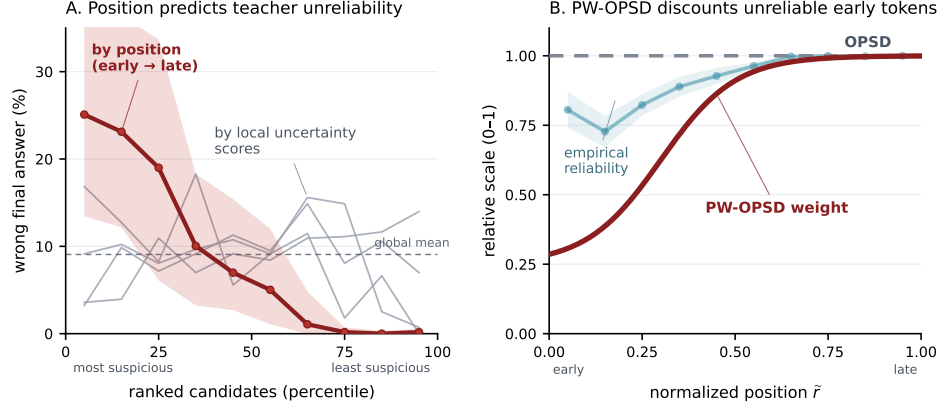


Figure 1: Branch viability reveals a positional reliability structure in Qwen3-4B reasoning traces. Alternatives are selected by the same model under the privileged-answer teacher prompt and rolled out under the ordinary student prompt; for high-ambiguity candidate positions, these forced student-context continuations fail more often at early positions and rarely fail late in the trace. **(A)** Sorting candidates by normalized position therefore separates teacher-unreliable branch points from viable diversity, whereas local uncertainty (entropy) scores yield near-flat failure curves. **(B)** The resulting empirical reliability curve motivates the PW-OPSD weight schedule, which discounts early high-ambiguity positions while preserving full-strength supervision later. See section 3.2 for the details of the diagnostic experiments.

privileged teacher target as equally useful at every student-visited prefix. Recent adaptive, relaxed, or gated distillation objectives have challenged this uniformity assumption from complementary angles: EOPD augments reverse-KL OPD with forward KL on high-teacher-entropy tokens [Jin et al., 2026], TODI mixes forward and reverse KL per token using a teacher-student probability log-ratio [Jung et al., 2025], REOPOLD treats the teacher-student log-likelihood ratio as a token reward with reward clipping and entropy-based sampling [Ko et al., 2026], and GATES gates privileged self-distillation by tutor consensus [Stein et al., 2026]. Together, these works motivate nonuniform token-level supervision through local entropy, teacher-student mismatch, clipped token rewards, or consensus signals. However, all of these adaptive criteria share a common limitation: they measure how ambiguous the teacher’s local distribution is, through entropy, teacher-student mismatch, or tutor consensus, but token-local ambiguity is *not* the same as low teacher reliability, and conflating the two reweights tokens for the wrong reason. Our point of departure is the reliability meaning of these local signals. Entropy-based criteria are a natural way to detect uncertainty, but high teacher entropy is not the same as low teacher reliability. A high-entropy teacher distribution can reflect non-viable uncertainty: several locally plausible next-token alternatives receive probability mass, but some of them fail to recover the correct answer under continued student-context generation. However, it can also reflect benign diversity: the teacher assigns mass to multiple viable solution routes or surface realizations that still preserve correctness. Thus a token-local uncertainty score can identify ambiguity, but it does not distinguish non-viable uncertainty from viable solution diversity.

The practical cost is that any reweighting following local ambiguity alone will either amplify supervision at branch points where the teacher derails, or downweight supervision at benign branch points where the teacher stays correct. Thus, the right object to estimate for adaptive token weighting is not teacher entropy but teacher *reliability*: a per-context probability that the teacher’s local target is useful supervision for the student. We formalize this *reliability* as a latent indicator I_t for whether matching the teacher target at distillation context c_t helps preserve or recover the correct solution. The reliability-weighted surrogate then weights the forward-KL term by $\rho^*(c_t) = \Pr(I_t = 1 \mid c_t)$.

To connect this latent quantity to observable behavior, we introduce a branch-viability diagnostic. Throughout, “teacher” and “student” refer to two prompt templates applied to the same model: the teacher template includes the privileged ground-truth answer; the student template is the ordinary problem prompt. Starting from a student-template rollout that reaches the correct answer, we use the teacher template to propose high-ambiguity next-token alternatives, force each alternative after the student-template prompt plus the spine prefix, and continue under the student template (no privileged information). We use student-template continuation because a teacher-template continuation can re-use the privileged answer and recover from almost any forced branch, collapsing the labels. If the

forced alternatives usually fail to recover the correct answer, the candidate is labeled *real-uncertain*: at that prefix, the teacher’s local alternatives are low-reliability supervision for the student distribution. If the alternatives remain viable, the ambiguity is treated as benign diversity rather than as evidence of low teacher reliability.

On Qwen3-4B [Yang et al., 2025], this diagnostic shows a clear positional pattern, as demonstrated in Fig. 1. In the correct-spine subset, real-uncertain candidates concentrate early in the normalized trace, while diversity candidates are more common later. After within-problem residualization, an oriented position score separates real-uncertain candidates from diversity candidates with AUROC 0.83 and a 95% cluster-bootstrap interval of [0.66, 0.95]; the tested local uncertainty diagnostics reach residual AUROC at most 0.57. Thus the diagnostic supplies the direction of the reliability prior: early high-ambiguity branch points are where the privileged teacher’s local alternatives most often become unreliable, while later positions are more reliable under forced continuation.

Inspired by this finding, we propose Position-Weighted On-Policy Self-Distillation (PW-OPSD), which implements this reliability trend as a plug-in approximation to $\rho^*(c_t)$. It keeps the same student rollout, privileged teacher pass, and per-vocabulary clipped forward-KL surrogate as OPSD, but aggregates token losses with an increasing sigmoid function of normalized within-sequence position. The floor keeps early tokens partially supervised, while the increasing schedule assigns stronger weight to later positions where the diagnostic indicates higher teacher reliability. The method therefore changes only the outer reliability-weighted aggregation, adding no extra teacher pass or auxiliary verifier.

We evaluate PW-OPSD on Qwen3-4B math reasoning benchmarks with the same maximum generation length of 38,912 tokens as the OPSD reference evaluation [Zhao et al., 2026]. The diagnostic-derived schedule matches OPSD on MATH-500 Avg@12 (95.34 vs. 95.33) and improves Avg@12 by +1.0 point on AIME 2024 and +1.1 points on AIME 2025, with the same teacher cost. The position-schedule sweep further shows how reliability-curve strength affects harder regimes: the aggressive schedule improves HMMT 2025 Avg@12 by +1.48 points over OPSD and improves Maj@12 by +1.11 points. These results support position as a structural reliability prior and show that reasoning distillation benefits from modeling teacher reliability along the trajectory.

2 Related Work

On-policy distillation. Knowledge distillation trains a student to match soft teacher probabilities rather than only hard labels [Hinton et al., 2015]. Sequence-level distillation extends this idea from token labels to generated completions [Kim and Rush, 2016]. Modern LLM post-training popularized instruction and feedback-based tuning [Ouyang et al., 2022]; in distillation, recent LLM methods often move supervision on policy, so the student is trained on prefixes it actually visits [Agarwal et al., 2023, Gu et al., 2023, Ko et al., 2024, Zhao et al., 2026]. This follows the interactive imitation-learning intuition of querying expert feedback on learner-visited states [Ross et al., 2010]; empirical work on LLM imitation also reports degradation under off-policy imitation in multi-step generation settings [Gudibande et al., 2023]. GKD formalizes generalized on-policy knowledge distillation [Agarwal et al., 2023], MiniLLM optimizes sequence-level reverse-KL distillation on student/teacher-mixed on-policy samples through a policy-gradient formulation [Gu et al., 2023], and DistiLLM studies skewed KL objectives for white-box OPD on student-generated outputs [Ko et al., 2024]. See Song and Zheng [2026] for a broader survey of OPD. OPSD is the main baseline in this paper: it uses a privileged teacher prompt containing reference information to provide token-level targets along student rollouts [Zhao et al., 2026]. PW-OPSD keeps the privileged on-policy construction of OPSD and makes the outer reliability weight trajectory-structured rather than uniform.

Divergence choice and adaptive token weighting. Several distillation methods change the divergence or token-level controller between teacher and student. MiniLLM replaces standard forward-KL KD with sequence-level reverse KL and optimizes the resulting on-policy objective via policy-gradient returns [Gu et al., 2023]; DistiLLM uses skewed KL objectives to stabilize white-box OPD on student-generated outputs [Ko et al., 2024]. Broader f -divergence work studies how divergence choice changes sequence-level KD behavior [Wen et al., 2023]. AKL revisits the usual mode-seeking/mode-covering framing for LLM distillation, arguing that forward and reverse KL share the same asymptotic objective but that FKL emphasizes head probabilities while RKL emphasizes tail probabilities in early epochs, then proposing an adaptive KL mixture [Wu et al., 2024]. Recent

adaptive or privileged distillation methods also depart from uniform token-level supervision, but for different reasons and with different control signals. EOPD applies forward KL on high-teacher-entropy tokens while retaining reverse KL elsewhere in OPD [Jin et al., 2026]; REOPOLD [Ko et al., 2026] interprets OPD as policy optimization with a teacher-student log-likelihood-ratio token reward and relaxes strict imitation through reward clipping, entropy-based dynamic sampling, and exploration-to-refinement training. Outside this student-rollout OPD divergence-switching line, TODI adaptively combines forward and reverse KL per token using a teacher-student probability log-ratio [Jung et al., 2025]; GATES instead gates privileged self-distillation using consensus among tutor traces [Stein et al., 2026]. These methods show that distillation supervision can be modulated per token through entropy, teacher-student ratios, clipped rewards, or consensus signals. However, these local control signals do not directly distinguish non-viable uncertainty from benign solution diversity. To explore this challenge, we propose a branch-viability experiment in which the same model, under the privileged-answer teacher prompt, selects forced alternatives and then rolls them out under the ordinary student prompt to test whether the correct answer is still recovered. We compare branch viability against standard uncertainty diagnostics, including predictive entropy, MC-dropout mutual information, and Dirichlet or evidential uncertainty [Kendall and Gal, 2017, Gal and Ghahramani, 2015, Malinin and Gales, 2018, Sensoy et al., 2018]. These baselines quantify local ambiguity, but they do not test whether high-probability teacher alternatives preserve correctness under continued generation.

3 Method

PW-OPSD is a reliability-weighted version of OPSD that preserves the student rollout, privileged teacher pass, and per-vocabulary clipped forward-KL surrogate. It changes the outer token aggregation through a position-dependent reliability weight and a per-sequence mean, motivated by a branch-viability diagnostic showing that early high-ambiguity branch points are often teacher-unreliable, while local uncertainty scores are weak predictors of this reliability event.

3.1 Background: On-Policy Self-Distillation

Let x_{stu} be the ordinary student prompt and x_{tch} be the privileged teacher prompt that includes the reference solution. On-policy self-distillation (OPSD) [Zhao et al., 2026] first samples a completion from the current student,

$$y_{1:T} \sim \pi_{\theta}(\cdot \mid x_{\text{stu}}),$$

and then queries a teacher copy of the same model on the privileged context and the student prefix. At each valid generated position t , the teacher target is

$$\mathbf{p}_t^{\text{opsd}} = p_{\text{tch}}(\cdot \mid x_{\text{tch}}, y_{<t}), \quad p_t = \pi_{\theta}(\cdot \mid x_{\text{stu}}, y_{<t}).$$

Both distributions are temperature-scaled before the softmax, matching the trainer path used for all forward-KL losses. With valid-token set $\mathcal{M}$, the implemented OPSD objective is

$$\mathcal{L}_{\text{OPSD}} = \frac{1}{|\mathcal{M}|} \sum_{t \in \mathcal{M}} \sum_{j \in \mathcal{V}} \min \left(\mathbf{p}_t^{\text{opsd}}(j) \log \frac{\mathbf{p}_t^{\text{opsd}}(j)}{p_t(j)}, \tau_{\text{clip}} \right). \quad (1)$$

The clipping in eq. (1) is element-wise over vocabulary terms: the quantity $\mathbf{p}_t^{\text{opsd}}(j) \log(\mathbf{p}_t^{\text{opsd}}(j)/p_t(j))$ is clipped for each $j \in \mathcal{V}$, the clipped terms are summed over the vocabulary, and the result is averaged over valid tokens. This objective is the uniform-weight baseline whose outer token aggregation PW-OPSD will modify.

3.2 Motivation: Token-Position Predicts Teacher Reliability

Branch-viability protocol. The branch-viability diagnostic asks whether a locally ambiguous teacher preference is reliable supervision for the student distribution. Throughout the diagnostic, “teacher” and “student” refer to two prompt templates applied to the same Qwen3-4B model: the *teacher* template includes the privileged ground-truth answer, while the *student* template is the ordinary problem prompt without that information.

For each problem, the model under the student template first generates an on-policy rollout; this fixed token sequence is the *spine*. The model is then re-evaluated along the spine under the teacher template,

Table 1: Branch-viability classification on Qwen3-4B. Entries are within-problem residualized AUROC for separating 8 *real-uncertain* candidates (whose forced student-context continuations fail to recover the correct answer) from 271 *diversity* candidates (whose continuations remain viable) across 61 problems, restricted to the correct-spine subset and to high-truncated-entropy candidate positions. For position, the scored feature is $1 - \tilde{r}$, so larger values correspond to earlier tokens. Cluster-bootstrap 95% CIs are computed over problems.

score (residualized within problem)	residual AUROC	95% CI
$H(\mathbf{p}_t^{\text{oprd}})$ (mean predictive entropy)	0.51	[0.50, 0.84]
MI (canonical MC-dropout epistemic)	0.51	[0.50, 0.62]
$\log \hat{\kappa}$ (Dirichlet precision)	0.54	[0.50, 0.69]
h_{trunc} (top-16 valid entropy)	0.57	[0.51, 0.72]
$1 - \tilde{r}$ (early-position score)	0.83	[0.66, 0.95]

which adds the privileged answer. Candidate positions are pre-selected from high top-16 valid-token truncated entropy on this teacher pass; Appendix A gives the filtering details. The diagnostic is deliberately targeted: it studies positions where the teacher already signals local ambiguity, rather than arbitrary positions on the rollout. The reliability-weighted view in section 3.3 identifies the latent reliability posterior as the formal object that this diagnostic probes behaviorally.

At each candidate position, the teacher proposes preferred next-token alternatives. We then test whether each alternative leads back to the correct answer *from the student’s perspective*: we form a prefix consisting of the student-template prompt, the spine truncated to the candidate position, and the forced alternative token, and complete the sequence under the student template (no privileged information). A candidate is labeled *real-uncertain* when these student-context continuations fail to recover the correct final answer; in this case the teacher’s local alternatives are unreliable targets for that student prefix. A candidate is labeled *diversity* when the alternatives remain viable routes to the same answer. We continue under the student template rather than the teacher template because student rollouts and deployment use the ordinary prompt without privileged information, and a teacher-template continuation can recover from almost any forced branch by re-using the ground-truth answer in its prompt; that recovery affordance would mask the very reliability question the diagnostic is asking. Appendix A gives the thresholded implementation details for the repeated forced-continuation rollouts.

Empirical finding. Table 1 reports 8 real-uncertain and 271 diversity candidates from 61 problems, restricted to the correct-spine subset; the underlying failure-rate-by-position curves are visualized in Fig. 1. Each feature is residualized within problem before scoring, so the AUROC reflects within-problem token-level separation rather than problem difficulty. Confidence intervals are cluster bootstraps over problems.

Position is the strongest predictor among the diagnostics we test. We define normalized position as $\tilde{r} = (\text{spine_pos} + 0.5)/L$, where L is the length of the student spine. Since the real-uncertain candidates concentrate early, table 1 reports the oriented early-position score $1 - \tilde{r}$ for the real-uncertain label. This score reaches AUROC 0.83 with 95% CI [0.66, 0.95], whereas predictive entropy, MC-dropout mutual information, Dirichlet precision, and top-16 truncated entropy remain weak predictors. Equivalently, raw position $\tilde{r}$ is negatively associated with teacher unreliability, which is why PW-OPSD uses an increasing reliability weight.

The local diagnostics in table 1 include standard teacher-side uncertainty scores: predictive entropy [Kendall and Gal, 2017], MC-dropout mutual information [Gal and Ghahramani, 2015], and Dirichlet precision and evidential-style categorical uncertainty [Malinin and Gales, 2018, Sensoy et al., 2018], plus our top-16 truncated entropy. They are included as diagnostic baselines for reliability prediction; implementation details for the MC-dropout scores are in Appendix B.

Implication for adaptive distillation signals. The weak scores in table 1 call into question a common assumption behind recent adaptive, relaxed, or privileged distillation methods: that local uncertainty or teacher–student discrepancy can serve as a reliable control signal for distillation. Existing methods instantiate this idea through teacher token entropy in EOPD [Jin et al., 2026], token-wise teacher–student probability log-ratios in ToDI [Jung et al., 2025], teacher–student log-likelihood-ratio rewards and entropy-guided token-level sampling in REOPOLD [Ko et al., 2026], and tutor-answer consensus in GATES [Stein et al., 2026]. These signals are useful for adapting

objectives, rewards, or trajectory selection, but our diagnostic shows that they are not reliable proxies for branch viability: the same large or permissive signal can arise when the teacher is unreliable and when multiple continuations are genuinely viable. Position provides a complementary structural signal, capturing where local ambiguity becomes unreliable for the student.

3.3 Theoretical Interpretation: Reliability-Weighted Distillation

Let $c_t = (x_{\text{stu}}, x_{\text{tch}}, y_{<t})$ denote the distillation context, let q_t be the privileged teacher target at that context, and let $p_t = \pi_\theta(\cdot \mid x_{\text{stu}}, y_{<t})$ be the student distribution. For the interpretation, define the unclipped token divergence $D_t(\theta) = \text{KL}(q_t \parallel p_t)$. Introduce a latent indicator $I_t \in \{0, 1\}$, where $I_t = 1$ means that the teacher’s local target is reliable for the student prefix: matching it helps preserve or recover the correct solution rather than following an off-path alternative. The ideal reliability-filtered risk is

$$R(\theta) = \mathbb{E}[I_t \cdot D_t(\theta)]. \quad (2)$$

For fixed θ , $D_t(\theta)$ is determined by the distillation context, so the tower property gives

$$R(\theta) = \mathbb{E}[\rho^*(c_t) \cdot D_t(\theta)], \quad \rho^*(c_t) = \Pr(I_t = 1 \mid c_t). \quad (3)$$

The Bayes-optimal surrogate for this latent-reliability risk weights each token by the posterior probability that the teacher target is reliable. Appendix C gives the conditioning details. On-policy and adaptive distillation methods exercise related levers through sampling distributions, divergences, or weight functionals [Agarwal et al., 2023, Gu et al., 2023, Ko et al., 2024, Wen et al., 2023, Wu et al., 2024, Jung et al., 2025, Jin et al., 2026]; PW-OPSD uses normalized position as a low-cost structural proxy for that posterior.

The branch issue can also be summarized by a mixture identity. Let $h_t = (x_{\text{stu}}, y_{<t})$ denote the student-side prefix. Suppose the teacher target at that prefix is a mixture over latent successful branches Z ,

$$q_t = \sum_z \alpha(z \mid h_t) q_t^z.$$

For any student distribution p_t ,

$$\mathbb{E}_{z \sim \alpha(\cdot \mid h_t)}[\text{KL}(q_t^z \parallel p_t)] = \text{KL}(q_t \parallel p_t) + I_q(Y_t; Z \mid h_t), \quad (4)$$

where $I_q(\cdot; \cdot \mid \cdot)$ is conditional mutual information under the teacher mixture. The mutual-information term quantifies branch-specific variation hidden by the marginal teacher target; because it is independent of p_t , the identity is not by itself a different gradient objective. The corresponding sequence-level identity accumulates this conditional mutual information across time; Appendix D gives the details and interpretation.

The branch-viability study supplies the task-specific direction: early high-ambiguity positions are most often labeled real-uncertain, so PW-OPSD discounts them with a lower outer weight.

3.4 PW-OPSD: Position-Weighted On-Policy Self-Distillation

PW-OPSD uses a deterministic structural proxy for the reliability posterior $\rho^*(c_t)$: a token’s relative position in its own student rollout. For sequence i with valid completion length L_i , define the one-based valid token index $t \in \{1, \dots, L_i\}$ and the per-row position fraction

$$r_{i,t} = \frac{t - 0.5}{L_i}.$$

The position weight is

$$w_{i,t} = w_{\min} + (1 - w_{\min}) \sigma\left(\frac{r_{i,t} - \tau}{s}\right), \quad (5)$$

with defaults $(w_{\min}, \tau, s) = (0.25, 0.30, 0.10)$. The floor keeps early tokens partially supervised, the threshold places the transition early in the rollout, and the scale avoids a hard discontinuity. The schedule is increasing because raw position is positively associated with reliability in the branch-viability diagnostic.

The per-row normalization in $r_{i,t}$ and the per-sequence reduction are a single design choice. A position fraction is meaningful only relative to the sequence’s own valid length, not the batch-padded

maximum. The corresponding loss therefore averages within each sequence before averaging across valid sequences; otherwise, longer rollouts would receive more gradient mass purely because they contain more tokens.

With B valid sequences in the batch, the implemented PW-OPSD objective is

$$\mathcal{L}_{\text{PW-OPSD}} = \frac{1}{B} \sum_{i=1}^B \frac{1}{L_i} \sum_{t=1}^{L_i} w_{i,t} \sum_{j \in \mathcal{V}} \min \left(\mathbf{p}_{i,t}^{\text{opsd}}(j) \log \frac{\mathbf{p}_{i,t}^{\text{opsd}}(j)}{\pi_{\theta}(j \mid x_{\text{stu}}^i, y_{<t}^i)}, \tau_{\text{clip}} \right). \quad (6)$$

The inner term is the same per-vocabulary clipped forward KL as eq. (1). PW-OPSD uses the ordinary single-pass teacher target $\mathbf{p}_{i,t}^{\text{opsd}}$; its additional computation over OPSD is only the scalar sigmoid in eq. (5).

4 Experiments

Section 3.2 suggests a simple reliability prior: early high-ambiguity branch points are more likely to be teacher-unreliable, while later positions are more reliable under the branch-viability diagnostic. This section tests whether the corresponding position-weighted objective improves downstream reasoning behavior relative to OPSD under the same rollout, teacher pass, and evaluation protocol.

4.1 Setup

Models, baselines, and schedule. We train on the Qwen3-4B checkpoint [Yang et al., 2025, Qwen Team, 2025] for the main comparison (table 2), the position-schedule sweep (table 3), and the reduction-positioning ablation (table 6); the cross-model evidence in table 4 additionally uses DeepSeek-R1-Distill-Llama-8B (DSR1-L8B) [Guo et al., 2025, DeepSeek-AI, 2025] and Olmo-3-7B-Think [Allen Institute for AI, 2026]. PW-OPSD uses the Moderate schedule $(w_{\min}, \tau, s) = (0.25, 0.30, 0.10)$, the diagnostic-derived default. We compare against three baselines: OPSD [Zhao et al., 2026] as the uniform-weight reference, EOPD [Jin et al., 2026] as a representative entropy-conditioned adaptive-KL baseline, and REOPOLD [Ko et al., 2026] as a cross-family policy-gradient adaptive on-policy distillation baseline that controls for whether any adaptive per-token signal recovers the downstream pattern attributed to position. All methods share the OPSD privileged on-policy chassis with LoRA [Hu et al., 2021] (rank 64, $\alpha = 128$), are evaluated at the 100-step checkpoint following OPSD, and otherwise use their published defaults. Full training hyperparameters, the cross-family rationale, and the PW-OPSD pseudocode are in Appendices G and I.

Evaluation. We follow the OPSD evaluation setting [Zhao et al., 2026] with maximum generation length 38,912 tokens: `max_new_tokens= 38912`, `val_n= 12` samples per problem, temperature $T = 1.0$, `top-p = 0.95`, `top-k` disabled, and `enable_thinking=True`. Benchmarks are MATH-500 [Hendrycks et al., 2021, Lightman et al., 2023, HuggingFaceH4, 2024], AIME 2024 [HuggingFaceH4, 2025], AIME 2025 [yentinglin, 2025], and HMMT 2025 [Dekoninck et al., 2026]. We report Pass@12, Avg@12, and Maj@12; Appendix J gives the formulas. For each method–benchmark pair we run three random evaluation seeds (main, 1, 2) and report mean $\pm$ across-seed sample standard deviation; the cross-model assessment (table 4) applies the same four-benchmark three-seed protocol to two additional checkpoints from different model families.

4.2 Main results on Qwen3-4B

Table 2 reports the Qwen3-4B comparison with maximum generation length 38,912 tokens. We report the diagnostic-derived *Moderate* schedule $(w_{\min}, \tau, s) = (0.25, 0.30, 0.10)$ as the default PW-OPSD configuration. We also include *Aggressive* (0.05, 0.50, 0.05) from the schedule sweep as a sensitivity variant that applies a stronger early discount. The main pattern is that position weighting improves per-sample reasoning accuracy on the AIME benchmarks while preserving near-saturated MATH-500 performance. On MATH-500, OPSD, EOPD, and PW-OPSD Moderate are tied within 0.01 pp Avg@12, while PW-OPSD Aggressive gives the highest MATH-500 Avg@12 and Maj@12 in the table. On AIME 2024 and AIME 2025, PW-OPSD Moderate improves Avg@12 over OPSD by +1.0 pp and +1.1 pp, respectively. The local or adaptive alternatives do not reproduce this Avg@12 pattern: EOPD trails OPSD by 1.6 pp on AIME 2024 and 1.0 pp on AIME 2025, while REOPOLD trails by 1.2 pp and 4.5 pp. HMMT 2025 highlights the role of reliability-curve strength.

Table 2: Qwen3-4B results with maximum generation length 38,912 tokens. Entries report mean $\pm$ standard deviation across three evaluation seeds. Bold marks the best value for each benchmark-metric column and any value within 0.1 pp of the leader. The Aggressive row is included to show schedule sensitivity rather than as a separate method claim.

Method	Benchmark	Pass@12	Avg@12	Maj@12
OPSD [Zhao et al., 2026]	MATH-500	98.00 $\pm$ 0.35	95.33 $\pm$ 0.08	96.73 $\pm$ 0.12
	AIME 2024	86.67 $\pm$ 0.00	75.19 $\pm$ 0.42	80.00 $\pm$ 0.00
	AIME 2025	83.33 $\pm$ 0.00	66.67 $\pm$ 1.27	73.33 $\pm$ 3.33
	HMMT 2025	64.44 $\pm$ 1.92	43.89 $\pm$ 0.73	51.11 $\pm$ 3.85
EOPD [Jin et al., 2026]	MATH-500	98.33 $\pm$ 0.23	95.33 $\pm$ 0.08	96.73 $\pm$ 0.23
	AIME 2024	85.56 $\pm$ 1.92	73.61 $\pm$ 2.17	80.00 $\pm$ 0.00
	AIME 2025	82.22 $\pm$ 1.92	65.65 $\pm$ 0.89	76.67 $\pm$ 3.33
	HMMT 2025	63.33 $\pm$ 3.33	41.94 $\pm$ 2.00	50.00 $\pm$ 3.33
REOPOLD [Ko et al., 2026]	MATH-500	98.40 $\pm$ 0.20	95.09 $\pm$ 0.09	96.53 $\pm$ 0.31
	AIME 2024	86.67 $\pm$ 0.00	73.98 $\pm$ 0.42	80.00 $\pm$ 3.33
	AIME 2025	81.11 $\pm$ 1.92	62.13 $\pm$ 1.67	74.44 $\pm$ 5.09
	HMMT 2025	58.89 $\pm$ 1.92	41.39 $\pm$ 1.44	51.11 $\pm$ 3.85
PW-OPSD Moderate (ours)	MATH-500	98.40 $\pm$ 0.00	95.34 $\pm$ 0.10	96.67 $\pm$ 0.12
	AIME 2024	85.56 $\pm$ 1.92	76.20 $\pm$ 0.58	80.00 $\pm$ 0.00
	AIME 2025	83.33 $\pm$ 0.00	67.78 $\pm$ 1.27	74.44 $\pm$ 1.92
	HMMT 2025	60.00 $\pm$ 3.33	43.33 $\pm$ 1.21	52.22 $\pm$ 1.92
PW-OPSD Aggressive (ours)	MATH-500	98.40 $\pm$ 0.20	95.53 $\pm$ 0.04	97.07 $\pm$ 0.12
	AIME 2024	87.78 $\pm$ 1.92	75.19 $\pm$ 0.85	80.00 $\pm$ 0.00
	AIME 2025	83.33 $\pm$ 0.00	67.59 $\pm$ 0.85	75.56 $\pm$ 1.92
	HMMT 2025	60.00 $\pm$ 0.00	45.37 $\pm$ 0.58	52.22 $\pm$ 1.92

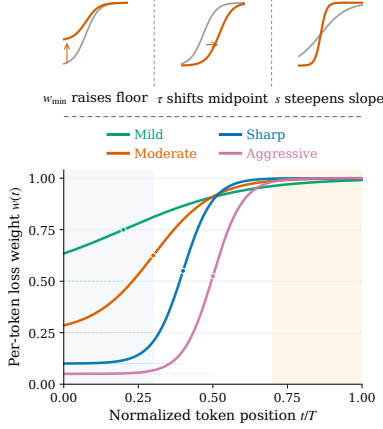
The Moderate schedule improves Maj@12 over OPSD by +1.11 pp but trails by 0.6 pp Avg@12 and 4.4 pp Pass@12. Under the stronger early discount of the Aggressive schedule, HMMT 2025 Avg@12 rises to 45.37 pp, a +1.48 pp gain over OPSD, and Maj@12 remains +1.11 pp above OPSD. Among the four evaluated methods we compare, PW-OPSD is the only one to improve over OPSD: Moderate gains +1.0 pp on AIME 2024 and +1.1 pp on AIME 2025, and Aggressive gains +1.5 pp on HMMT 2025. EOPD and REOPOLD instead regress on three of four benchmarks, losing 1.1 and 2.1 pp Avg@12 on average.

4.3 Position-schedule ablation

The main comparison fixes the PW-OPSD schedule to $(w_{\min}, \tau, s) = (0.25, 0.30, 0.10)$. Since the method introduces three scalar parameters, we also run a full schedule sweep over four settings to test whether the effect is tied to a single parameter tuple. Figure 2 shows the schedule shapes; Table 3 reports the completed Qwen3-4B ablation evaluations across all four configurations under the same maximum generation length as the main table.

The four settings vary how strongly early tokens are discounted. Mild (0.50, 0.20, 0.20) keeps a high early floor and uses the softest transition; Moderate (0.25, 0.30, 0.10) is the chosen configuration from Table 2; Sharp (0.10, 0.40, 0.05) lowers the floor and makes the transition steeper; Aggressive (0.05, 0.50, 0.05) applies the strongest early down-weighting and delays the transition furthest. Thus lower $w_{\min}$ makes the early-token discount more aggressive, larger τ moves the transition later, and smaller s makes it sharper.

On MATH-500, every configuration sits within 0.2 pp Avg@12, well inside across-seed noise. On AIME 2024 Moderate holds the Avg@12 lead but the other three trail by at most 2.2 pp and all four tie on Maj@12; AIME 2025 Avg@12 is similarly clustered (67.59–68.80). HMMT 2025 separates the configurations more clearly: Aggressive and Mild gain +1.9–2.0 pp Avg@12 over Moderate. No configuration dominates uniformly. We retain Moderate in the main table because it is the schedule chosen *a priori* from the diagnostic curves, not because it is the best across this sweep. Averaged across the four benchmarks, all four schedules improve Avg@12 over OPSD (70.27) by +0.30 to +0.65 pp and span only 0.35 pp among themselves, confirming PW-OPSD is robust to the choice of $(w_{\min}, \tau, s)$.



Config	Benchmark	Pass@12	Avg@12	Maj@12
Mild	MATH-500	98.33 ± 0.12	95.44 ± 0.05	96.87 ± 0.12
	AIME 2024	87.78 ± 3.85	73.98 ± 0.85	80.00 ± 0.00
	AIME 2025	84.44 ± 1.92	67.59 ± 1.16	73.33 ± 0.00
	HMMT 2025	62.22 ± 1.92	45.28 ± 0.73	47.78 ± 1.92
Moderate	MATH-500	98.40 ± 0.00	95.34 ± 0.10	96.67 ± 0.12
	AIME 2024	85.56 ± 1.92	76.20 ± 0.58	80.00 ± 0.00
	AIME 2025	83.33 ± 0.00	67.78 ± 1.27	74.44 ± 1.92
	HMMT 2025	60.00 ± 3.33	43.33 ± 1.21	52.22 ± 1.92
Sharp	MATH-500	98.13 ± 0.31	95.36 ± 0.15	96.60 ± 0.20
	AIME 2024	88.89 ± 7.70	75.19 ± 1.53	80.00 ± 0.00
	AIME 2025	83.33 ± 0.00	68.80 ± 0.89	75.56 ± 3.85
	HMMT 2025	64.44 ± 1.92	44.26 ± 1.12	46.67 ± 3.33
Aggressive	MATH-500	98.40 ± 0.20	95.53 ± 0.04	97.07 ± 0.12
	AIME 2024	87.78 ± 1.92	75.19 ± 0.85	80.00 ± 0.00
	AIME 2025	83.33 ± 0.00	67.59 ± 0.85	75.56 ± 1.92
	HMMT 2025	60.00 ± 0.00	45.37 ± 0.58	52.22 ± 1.92

Figure 2: Per-token weight schedules for the four configurations; top panels label the role of each ($w_{\min}$, τ , s) knob. **Table 3:** Position-schedule sweep on Qwen3-4B (maximum generation length 38,912 tokens). Each cell is mean $\pm$ standard deviation across three evaluation seeds. Bold marks the best value for each benchmark-metric column within the sweep. Configurations: Mild (0.50, 0.20, 0.20), Moderate (0.25, 0.30, 0.10), Sharp (0.10, 0.40, 0.05), Aggressive (0.05, 0.50, 0.05).

Table 4: Cross-model Avg@12 evidence for PW-OPSD (Moderate, no per-model retuning) versus the OPSD baseline of Zhao et al. [2026] across three model families with maximum generation length 38,912 tokens. Entries report mean $\pm$ across-seed sample standard deviation over three evaluation seeds; the Avg@12 column is the equal-weight mean across the four benchmarks, computed from unrounded seed-level values. Bold marks the within-model winner in each numeric column, including values within 0.1 pp of the leader.

Model	Method	MATH-500	AIME 2024	AIME 2025	HMMT 2025	Avg@12
Qwen3-4B	OPSD	95.33 ± 0.08	75.19 ± 0.42	66.67 ± 1.27	43.89 ± 0.73	70.27
	PW-OPSD (ours)	95.34 ± 0.10	76.20 ± 0.58	67.78 ± 1.27	43.33 ± 1.21	70.66
DSR1-L8B	OPSD	89.02 ± 0.19	40.74 ± 2.58	31.57 ± 1.28	20.93 ± 0.80	45.56
	PW-OPSD (ours)	88.72 ± 0.25	41.20 ± 0.32	32.22 ± 0.56	21.48 ± 0.42	45.91
Olmo-3-7B-Think	OPSD	96.20 ± 0.08	72.78 ± 0.28	67.41 ± 2.16	43.89 ± 0.73	70.07
	PW-OPSD (ours)	96.26 ± 0.19	72.59 ± 1.53	69.63 ± 1.12	43.80 ± 0.58	70.57

Reduction vs. positioning. PW-OPSD differs from OPSD along two axes: a position-dependent token weight w_t (positioning) and a per-rollout average of the token loss before the batch mean (per-sequence reduction). The full 2×2 factorial in Table 6 (Appendix L) shows the joint configuration is the only one that matches the AIME 2024 lead of Table 2 (+1.0 pp Avg@12 over OPSD).

4.4 Cross-model transfer of the position schedule

To test whether the diagnostic-derived Moderate schedule transfers beyond Qwen3-4B, we evaluate PW-OPSD on two larger models from different families: DeepSeek-R1-Distill-Llama-8B (8B, Llama family) [Guo et al., 2025, DeepSeek-AI, 2025] and Olmo-3-7B-Think (7B, OLMo family) [Allen Institute for AI, 2026]. For each model we compare PW-OPSD (Moderate, ($w_{\min}$, τ , s) = (0.25, 0.30, 0.10)) against the OPSD baseline under the same training and evaluation protocols as the Qwen3-4B main experiments: the full four-benchmark suite (MATH-500, AIME 2024, AIME 2025, HMMT 2025), three independent evaluation seeds per cell, and the maximum generation length of 38,912 tokens described in section 4.1. The Moderate schedule is held fixed across all three models with no per-model retuning. For DSR1-L8B and Olmo-3-7B-Think, both trained as reasoning models, we additionally apply a tokenizer-level `<think>`-closure that replicates `enable_thinking=False` on chat templates that ignore the flag, matching the training-time student prompt format used on Qwen3-4B (evaluation continues to use `enable_thinking=True` as described in section 4.1).

Table 4 shows that the same Moderate schedule gives positive Avg@12 gains on every tested model: +0.39 pp on Qwen3-4B (4B parameters, Qwen family), +0.35 pp on DeepSeek-R1-Distill-Llama-8B (8B, Llama family), and +0.50 pp on Olmo-3-7B-Think (7B, OLMo family). The pattern spans

Qwen, Llama, and OLMo checkpoints across a 4B–8B parameter range, with no per-model schedule retuning, supporting the model-family portability of the position-rank reliability signal identified in section 3.2.

5 Conclusion, Limitations, and Future Work

In this paper, we study teacher-token reliability in on-policy self-distillation. To address whether high teacher entropy reflects low reliability or benign solution diversity, we design a branch-viability diagnostic, showing empirically that reliability is positionally structured (AUROC 0.83 vs. ≤ 0.57 for local uncertainty). Inspired by this finding, we introduce PW-OPSD, an increasing-sigmoid position weight on the OPSD chassis. Our experiments show that PW-OPSD improves Avg@12 over OPSD on AIME 2024 (+1.0 pp), AIME 2025 (+1.1 pp), and on DeepSeek-R1-Distill-Llama-8B and Olmo-3-7B-Think. One limitation is that PW-OPSD is plug-and-play yet its gains over OPSD remain modest, leaving room to design more sophisticated methods built on the positional reliability finding. To improve the proposed method, we aim to design more dedicated objectives that go beyond pure position weighting, such as position-conditioned mixing of forward and reverse KL.

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A Branch-viability protocol details

Models, prompts, and software. The diagnostic uses one Qwen/Qwen3-4B checkpoint [Qwen Team, 2025] (HuggingFace snapshot 1cfa9a72...3b3df60c, dtype bf16) with no adapters loaded. “Teacher” and “student” denote two prompt templates applied to this single checkpoint: the teacher template includes the privileged ground-truth answer, and the student template is the ordinary problem prompt. vLLM and HuggingFace load this checkpoint as separate software backends, not as different trained weights. Problems are drawn from three sources, sampled without replacement (see attrition below): MATH-500 test split [Hendrycks et al., 2021, Lightman et al., 2023, HuggingFaceH4, 2024], AIME 2024 [HuggingFaceH4, 2025], and AIME 2025 [yentinglin, 2025]. We use vllm 0.11.0 with tensor_parallel_size=4 (TP= 4) for generation [Kwon et al., 2023], transformers 4.57.1 for the HuggingFace (HF) teacher forward passes [Wolf et al., 2020], and torch 2.8.0+cu128 on 4×H100 80GB GPUs [Paszke et al., 2019].

Problem attrition. Phase A samples 24 MATH-500, 30 AIME 2024, and 30 AIME 2025 problems (84 total). Phase B produces $23 + 21 + 18 = 62$ correct-spine problems. After Phase F labeling, one AIME 2024 correct-spine problem has only gray candidates, so the binary-labeled pool used in table 1 contains $23 + 20 + 18 = 61$ usable problems. Phase C then proposes up to 5 high-truncated-entropy candidates per problem (with the spacing / plausibility filters below); Phase E rolls out 6 student continuations per (candidate, alternative) pair. The table 1 report restricts to high-truncated-entropy candidate positions and to the 8 candidates labeled *real-uncertain* plus 271 *diversity* candidates across these 61 problems. Figure 1 uses the broader continuous-viability subset before binary thresholding.

Random seeds. Phase A fallback sampling, Phase E forced continuations, and the $\tilde{r}$ residualization use vLLM seed 0; Phase D MC-dropout masks use a per-position random state derived from the model’s default generator. The cluster bootstrap in Phase G uses NumPy seed 0.

Score residualization and AUROC. Within each problem p , every raw uncertainty score u (or its sign-oriented form for position) is mean-centered: $u' = u - \bar{u}_p$. We compute AUROC on the residualized scores against the binary label defined in Phase F, using the standard Mann-Whitney U formula with mid-rank tie handling. The cluster bootstrap resamples problems (not candidates) with replacement and recomputes the within-problem residualization on each resample, giving the 95% intervals reported in table 1.

- **Phase A.** vLLM (TP=4) student greedy decode of 84 problems in total (24 from MATH-500, 30 from AIME 2024, and 30 from AIME 2025; the protocol is run independently per dataset and the labeled candidates are pooled in Phase G), with fallback to $T=0.7$ for problems where greedy does not reach \boxed within 16K tokens.
- **Phase B.** HF forward of the same checkpoint under the teacher template (privileged-answer prompt + spine), giving per-position teacher distributions and top- $M=16$ valid-token truncated entropy.
- **Phase C.** Apply junk / EOS / special filters, plausibility filter $p_2 \geq 0.02 \wedge p_2/p_1 \geq 0.10$, 64-token spacing constraint, and skip post-\boxed positions. Keep up to 5 high-truncated-entropy candidate positions per problem; for each, record the teacher pass’s top-3 valid child tokens as forced alternatives.
- **Phase D.** HF teacher MC dropout ($M=5$, $p=0.1$, last 4 layers) at each candidate; record H_{full} , MI, and $\log \hat{\kappa}$.
- **Phase E.** vLLM (TP=4) *student*-context forced continuation: each forced child token from Phase C is appended to the student-template prompt + spine truncated to the candidate position, and rollouts continue under the student template (no privileged information). We use student-template continuation rather than teacher-template continuation because a teacher-template continuation can re-use the privileged answer in its prompt and recover from almost any forced child, collapsing the labels toward diversity. Across the three datasets, up to $(24 + 30 + 30) \times 5 \times 3 \times 6 = 7560$ attempted rollouts at $T=1.0$, top_p= 0.95, with max_tokens dynamically clipped per request to fit max_model_len= 32K.
- **Phase F.** For each forced child, define viability as the fraction of its 6 student-context continuations whose extracted boxed answer matches the ground truth. Label a candidate *diversity* if at least two of its children have viability $\geq V_{\text{high}}=0.75$; label it *real-uncertain*

if every child has viability $< V_{\text{low}}=0.40$ and the mean child viability is $< V_{\text{low}}$; otherwise label “gray” and exclude from the AUROC.

- **Phase G.** AUROC + area-under-precision-recall-curve (AUPRC) + cluster-bootstrap by problem (2000 resamples, multiplicity-preserved); scatter and histograms.

Privileged-info teacher prompt. We use the following template (wrapped in the Qwen3 chat template with `enable_thinking=True`):

You are a privileged teacher. Solve the problem and end your solution with the correct boxed answer.
Problem: {problem}
Privileged ground-truth final answer: {answer}
Use the ground-truth answer above. Produce a complete step-by-step solution that ends with \boxed{{answer}}.

This is a simplified privileged-info injection in the spirit of OPSD’s training-time teacher prompt; it is not pixel-identical to the OPSD official template but conditions the teacher’s autoregressive distribution on the ground-truth answer for the duration of generation.

B MC-dropout diagnostic implementation

MC dropout is diagnostic-only in this paper. It is used to compute H_{full} , MI, and $\log \hat{\kappa}$ scores for table 1; it is not used to construct a training target or a token weight. All training methods use the ordinary single-pass teacher target $\mathbf{p}_t^{\text{opsd}}$.

MC dropout via forward hooks. Modern Qwen / Llama transformer blocks do not expose an `nn.Dropout` submodule. We inject MC dropout at inference time via forward hooks on the last $L = 4$ transformer layers; the hook applies `F.dropout(h, p, training=True)` to the layer output. We use $p = 0.1$ throughout and $M = 5$ MC samples. The unperturbed privileged-teacher forward provides $\mathbf{p}_t^{\text{opsd}}$; the perturbed forwards are used only for the diagnostic uncertainty scores. Hooks are removed in a `try/finally` block to prevent leaking dropout into subsequent forward passes.

Numerical stability of $\hat{\kappa}_t$. The moment-matching estimator for $\hat{\kappa}_t$ can produce negative values when the sample variance trace exceeds the categorical variance trace $1 - \|\bar{\mathbf{p}}_t\|_2^2$ (an artifact of small M). For table 1, we do not apply any training-style floor or log-normalization. We store the raw $\hat{\kappa}_t$ estimate and compute $\log \max(\hat{\kappa}_t, \epsilon)$ only at score time, with $\epsilon = 10^{-6}$ for the log transform.

Tail-bucket aggregation. For very large vocabularies (Qwen3 vocab $\approx 152\text{K}$), computing the full softmax per MC sample at every position is memory-bound. We compute moment statistics token-by-token and aggregate $\sum_m \|\mathbf{p}_t^{(m)}\|_2^2$ incrementally to avoid materializing the full per-sample distribution beyond one forward at a time. In Phase B of the branch-viability experiment we still materialize the per-position softmax once for top- M selection; this is a one-time pass per problem.

C Reliability-weighted surrogate details

Section 3.3 conditions the reliability posterior on the distillation context $c_t = (x_{\text{stu}}, x_{\text{tch}}, y_{<t})$. This conditioning makes the tower-property step explicit. For fixed θ and a fixed teacher, the token divergence $D_t(\theta) = \text{KL}(q_t \| p_t)$ is determined by c_t , so

$$\mathbb{E}[I_t D_t(\theta)] = \mathbb{E}[\mathbb{E}[I_t D_t(\theta) | c_t]] = \mathbb{E}[D_t(\theta) \Pr(I_t = 1 | c_t)].$$

Conditioning only on the student prefix $h_t = (x_{\text{stu}}, y_{<t})$ would require an additional assumption, because the privileged teacher target can vary with the teacher prompt and reference information even when the student prefix is fixed.

The risk in eq. (2) is also a distillation surrogate on sampled rollouts. As in standard OPSD implementations, the rollout is sampled from the current student and then treated as a fixed training example for the KL update; gradients pass through the student probabilities on visited prefixes, not through the sampling operation that produced those prefixes. The main-text interpretation uses

the unclipped divergence $D_t(\theta)$ for notational clarity, while the implemented objectives use the element-wise clipped forward-KL surrogate described in eq. (1).

D Branch-mixture identity and interpretation

Equation (4) gives the token-level identity used in the main text. With joint distribution $q(z, y_t | h_t) = \alpha(z | h_t) q_t^z(y_t)$ and marginal $q_t(y_t) = \sum_z \alpha(z | h_t) q_t^z(y_t)$,

$$\begin{aligned} \mathbb{E}_{z \sim \alpha} \text{KL}(q_t^z \| p_t) &= \sum_{z, y_t} q(z, y_t | h_t) \log \frac{q_t^z(y_t)}{p_t(y_t)} \\ &= \text{KL}(q_t \| p_t) + I_q(Y_t; Z | h_t). \end{aligned}$$

The mutual-information term is independent of p_t . The identity therefore decomposes branch-specific variation hidden by the marginal teacher target; it is not, by itself, a proof that marginal forward KL has a different gradient objective. The empirical branch-viability diagnostic supplies the task-specific direction used by PW-OPSD: early high-ambiguity positions more often correspond to unreliable supervision.

The same branch-mixture view yields a sequence-level form. Let the teacher’s branch-conditioned next-token distribution be $q_t^z(y_t | y_{<t})$, and let the corresponding sequence distribution factorise autoregressively as

$$q^z(y_{1:T}) = \prod_{t=1}^T q_t^z(y_t | y_{<t}).$$

Its marginal under the latent branch prior α is $q(y_{1:T}) = \sum_z \alpha(z) q^z(y_{1:T})$. For a student sequence distribution $p(y_{1:T}) = \prod_{t=1}^T p_t(y_t | y_{<t})$,

$$\mathbb{E}_{z \sim \alpha} [\text{KL}(q^z \| p)] = \text{KL}(q \| p) + \sum_t I_q(Y_t; Z | Y_{<t}). \quad (7)$$

Thus the value of branch-conditioned sequence forward KL differs from marginal sequence forward KL by the sum of conditional mutual-information terms between the next token and the latent branch. As in the token-level identity, this sum quantifies branch ambiguity but is independent of the student distribution.

E Position-schedule hyperparameters

The Moderate schedule used in the main table sets $(w_{\min}, \tau, s) = (0.25, 0.30, 0.10)$, placing the transition near the empirical early-to-late change in the branch-viability diagnostic while keeping a nonzero floor on early-token supervision. The full position sweep in table 3 covers four schedules: **Mild** (0.50, 0.20, 0.20), **Moderate** (0.25, 0.30, 0.10) (the a-priori choice from the diagnostic curve in fig. 1), **Sharp** (0.10, 0.40, 0.05), and **Aggressive** (0.05, 0.50, 0.05). The main table reports both Moderate (the a-priori headline) and Aggressive (the sweep configuration that recovers the HMMT 2025 cell on Avg@12). We did not tune $(w_{\min}, \tau, s)$ on any evaluation benchmark; the four configurations were chosen before observing any of the Avg@12/Pass@12/Maj@12 values.

F Adaptive-loss template and method comparison

Table 5 compares the three per-token distribution-matching objectives evaluated in table 2 along the dimensions that the template makes explicit: target distribution, adaptive signal or reliability proxy, and reduction over tokens. REOPOLD [Ko et al., 2026] is also evaluated in table 2; it does not fit this template because its gradient flows through the rolled-out token’s log-prob rather than through a distributional forward KL. Its role in the comparison is summarized in Appendix F.1.

F.1 Comparison to Related Adaptive Losses

Algorithm 1 (Appendix G) gives the explicit one-step pseudocode. For reference, the scalar-weighted clipped-FKL template is

$$\mathcal{L}_{\text{wFKL}} = \frac{1}{|\mathcal{M}|} \sum_{t \in \mathcal{M}} w_t \sum_{j \in \mathcal{V}} \min \left(\mathbf{p}_t(j) \log \frac{\mathbf{p}_t(j)}{\pi_\theta(j \mid x_{\text{stu}}, y_{<t})}, \tau_{\text{clip}} \right). \quad (8)$$

Different adaptive losses instantiate or modify this template through the choice of target, reliability proxy, inner divergence, and reduction.

OPSD uses a uniform weight [Zhao et al., 2026]. EOPD [Jin et al., 2026] gates a forward-KL augmentation to reverse KL using teacher entropy. PW-OPSD keeps the OPSD forward-KL inner loss and uses the position schedule $w_{i,t}$ from eq. (5) in its outer token aggregation. REOPOLD is also included in the experimental comparison but is a policy-gradient distillation variant rather than a per-token forward-KL weighting rule [Ko et al., 2026], so it does not instantiate the template above; section 4.1 states the cross-family rationale. Table 5 (Appendix F) lays out the three per-token distribution-matching objectives in a side-by-side table.

Table 5: Per-token distribution-matching distillation objectives compared in table 2, parametrized by target distribution, adaptive signal or reliability proxy, and reduction over tokens. REOPOLD is also evaluated in table 2 but follows a policy-gradient route rather than a per-token forward-KL weighting, so it is not instantiated in this template.

Method	target / inner loss	adaptive signal / proxy	reduction
OPSD [Zhao et al., 2026]	clipped FKL to $\mathbf{p}_t^{\text{opsd}}$	uniform ($w_t = 1$)	global token mean
EOPD [Jin et al., 2026]	entropy-gated RKL/FKL mixture	teacher-entropy gate	global token mean
PW-OPSD (ours)	clipped FKL to $\mathbf{p}_{i,t}^{\text{opsd}}$	position $w_{i,t}$	per-sequence mean

G PW-OPSD training pseudocode

Algorithm 1 gives one training step of PW-OPSD. The procedure computes student and teacher log-probabilities at the distillation temperature, forms the forward-KL tensor with no reduction, clamps each vocabulary element, sums over the vocabulary, applies the position weight, averages over valid tokens within each sequence, and then averages over valid sequences. The sampled rollout is fixed for this update, as discussed in Appendix C.

Algorithm 1 One training step of PW-OPSD.

Require: Parameters θ , prompts $\{(x_{\text{stu}}^i, x_{\text{tch}}^i)\}_{i=1}^B$, schedule $(w_{\min}, \tau, s)$, distillation temperature T_{distill} , clip τ_{clip} .

- 1: Sample student rollouts $y^i \sim \pi_\theta(\cdot \mid x_{\text{stu}}^i)$.
- 2: Let L_i be the number of valid generated tokens before EOS or truncation.
- 3: Score visited prefixes at T_{distill} under the privileged teacher context and ordinary student context to obtain $\mathbf{p}_{i,t}^{\text{opsd}}$ and $p_{i,t}$ for each valid t .
- 4: **for** each sequence i and valid token $t \in \{1, \dots, L_i\}$ **do**
- 5: $r_{i,t} \leftarrow (t - 0.5) / L_i$
- 6: $w_{i,t} \leftarrow w_{\min} + (1 - w_{\min}) \sigma((r_{i,t} - \tau) / s)$
- 7: $\ell_{i,t} \leftarrow \sum_{j \in \mathcal{V}} \min \left(\mathbf{p}_{i,t}^{\text{opsd}}(j) \log \frac{\mathbf{p}_{i,t}^{\text{opsd}}(j)}{p_{i,t}(j)}, \tau_{\text{clip}} \right)$
- 8: **end for**
- 9: $\mathcal{L}_{\text{PW-OPSD}} \leftarrow B^{-1} \sum_{i=1}^B L_i^{-1} \sum_{t=1}^{L_i} w_{i,t} \ell_{i,t}$
- 10: Update θ using $\nabla_\theta \mathcal{L}_{\text{PW-OPSD}}$.

H Implementation conventions inherited from OPSD

The following implementation conventions are held fixed across the methods reported in this paper.

Right-padded prompt collator + batch-max loss slicing. The upstream OPSD data_collator right-pads prompts and the trainer slices loss only on the first batch_max_prompt_len tokens of each completion; this is held constant across all methods in table 2. OpenThoughts-Math-30k [siyanzhao, 2026] prompts vary in length (median 93 tokens, max 826), so right-padding produces a per-batch prompt-PAD gap. Methods are compared on the same gap.

Train/eval prompt template gap. Training prompts use the OPSD reference student template Problem: {problem}\n\n Please reason... with enable_thinking=False; evaluation prompts use the simpler form {problem}\n\n Please reason... with enable_thinking=True. This training/evaluation prompt-template difference is shared across all methods reported in this paper.

Per-vocabulary clip semantics. The clipped forward KL in eq. (1) clamps each token/vocabulary entry of $q_t(j) \log(q_t(j)/p_t(j))$ element-wise via `F.kl_div(reduction='none').clamp(max=tau_clip)` before the inner sum over the vocabulary. This matches the OPSD reference implementation.

Gradient-accumulation token-mean. With per-microbatch token-mean reduction and gradient accumulation = 2, the OPSD loss computes the mean of two per-microbatch token-means rather than the exact token-weighted mean over the effective batch. This is the standard HuggingFace Trainer behavior and is held fixed for the OPSD baseline. PW-OPSD’s per-sequence reduction does not have this issue because each microbatch contains the same number of sequences.

vLLM rollout seed. The vLLM colocate-mode rollout sampler is seeded as `accelerator.process_index // tp_size`, independent of the trainer -seed flag. Re-runs of the same -seed N therefore produce slightly different rollouts. All evaluation results use seeded vLLM SamplingParams.seed, which is reported per evaluation run.

I Evaluation setup

Models. The Qwen3-4B checkpoint [Yang et al., 2025, Qwen Team, 2025] is used in the main comparison (table 2), the position-schedule sweep (table 3), and the reduction-positioning ablation (table 6); the cross-model evidence in table 4 additionally uses DeepSeek-R1-Distill-Llama-8B [Guo et al., 2025, DeepSeek-AI, 2025] and Olmo-3-7B-Think [Allen Institute for AI, 2026], two larger models from different families. In every comparison we train LoRA adapters on a fixed checkpoint and merge the selected adapter before evaluation, so within each model block, differences between method rows come only from the distillation objective and not from the checkpoint weights.

Training. We use the OPSD privileged on-policy setup [Zhao et al., 2026]. Across methods, the local implementation holds fixed preprocessing and prompt templates, full-vocabulary clipped forward-KL conventions, LoRA [Hu et al., 2021] rank 64, $\alpha = 128$, dropout 0.05, learning rate 5×10^{-6} , max_completion_length= 1024, distillation temperature $T_{\text{distill}} = 1.1$, KL clip $\tau_{\text{clip}} = 0.05$, a fixed teacher with LoRA disabled in the privileged forward pass, and seed 42. The local launcher uses $4 \times \text{H100}$ GPUs with effective batch size 32 (per-device batch 4 with gradient accumulation 2). All methods are evaluated at the 100-step checkpoint, following the OPSD evaluation horizon of Zhao et al. [2026]. PW-OPSD uses $(w_{\min}, \tau, s) = (0.25, 0.30, 0.10)$ for the diagnostic-derived default schedule.

Baselines. We compare PW-OPSD against three baselines run under our common training and evaluation protocol: OPSD [Zhao et al., 2026] as the uniform-weight reference, EOPD [Jin et al., 2026] as a representative entropy-conditioned adaptive-KL alternative, and REOPOLD [Ko et al., 2026] as a representative policy-gradient adaptive on-policy distillation method. REOPOLD differs from the per-token forward-KL family in gradient form, because its gradient flows through the rolled-out token’s log-prob rather than through a distributional forward KL. It nevertheless answers the same operational question of how to derive a per-token training signal from teacher-student log-likelihood-ratio rewards on sampled rollout tokens. Including REOPOLD therefore provides a cross-family reference point that controls for the possibility that any adaptive token weighting

recovers the downstream pattern attributed to position. All baselines use their published default hyperparameters under the common training and evaluation protocol described above.

Evaluation setting. We use a maximum generation length of 38,912 tokens: `max_new_tokens=38912`, $N = 12$ samples per problem, temperature $T = 1.0$, `top-p = 0.95`, `top-k` disabled, and `enable_thinking=True`, matching the OPSD evaluation setup [Zhao et al., 2026].

Metrics. We report Avg@12, Pass@12, and Maj@12, which together expose per-sample accuracy, search-style success under repeated attempts, and stability under aggregation. Majority vote uses math-equivalence clustering with `sympy` [Meurer et al., 2017] and `math_verify` [Kydliček, 2025]; unformatted predictions are placed in a single INVALID cluster that participates in the plurality count and is scored incorrect when selected. Formal per-problem definitions are deferred to Appendix J.

Benchmarks. We evaluate on MATH-500 [Hendrycks et al., 2021, Lightman et al., 2023, HuggingFaceH4, 2024], AIME 2024 [HuggingFaceH4, 2025], AIME 2025 [yentinglin, 2025], and HMMT February 2025. The HMMT set is a locally cleaned parquet derived from MathArena’s `hmmt_feb_2025` release [Dekoninck et al., 2026], with SHA-256 recorded in the appendix. For each method–benchmark pair we run three random evaluation seeds (main, 1, 2) and report mean $\pm$ across-seed sample standard deviation. The cross-model assessment (table 4) extends this protocol to DeepSeek-R1-Distill-Llama-8B and Olmo-3-7B-Think under the same four benchmarks.

J Evaluation metric definitions

Multi-sample evaluation for reasoning. Reasoning models are commonly evaluated with repeated sampling because a single completion can understate the chance that the model finds a correct solution. Pass@N measures whether any of N samples succeeds [Chen et al., 2021, Brown et al., 2024], while self-consistency and majority vote aggregate multiple reasoning paths [Wang et al., 2022]. We report Avg@12, Pass@12, and Maj@12 throughout. These metrics expose different behaviors of a distillation objective: per-sample accuracy, search-style success under repeated attempts, and stability under aggregation.

Formal definitions. For a single problem with $N=12$ generated solutions $\{y^{(1)}, \dots, y^{(N)}\}$, the predicted answer is extracted as the content of the last `\boxed{...}` in $y^{(i)}$; samples without a parseable boxed answer are assigned the cluster key INVALID. Let $c_i \in \{0, 1\}$ indicate whether $y^{(i)}$ is graded correct against the gold answer (using `math_verify` [Kydliček, 2025] with a normalized string-equality fallback for parsing failures), and let a_i be the math-equivalence cluster key obtained by grouping the extracted answers under the `sympy` [Meurer et al., 2017] and `math_verify` normalization pipeline. The per-problem metrics are

$$\text{Avg@}N = \frac{1}{N} \sum_{i=1}^N c_i, \quad (9)$$

$$\text{Pass@}N = \mathbf{1} \left[\sum_{i=1}^N c_i \geq 1 \right], \quad (10)$$

$$\text{Maj@}N = \begin{cases} 0, & a^* = \text{INVALID}, \\ c_{i^*}, & \text{otherwise,} \end{cases} \quad (11)$$

where the plurality cluster key is $a^* \in \arg \max_{a \in \{a_1, \dots, a_N\}} \#\{j : a_j = a\}$, with ties broken by smallest first occurrence $a^* = a_{\min\{i : a_i \in \arg \max\}}$, and the representative index $i^* = \min\{i : a_i = a^*\}$. The INVALID cluster *participates in the plurality count* but is scored incorrect when selected, so a problem on which the model produces no parseable answer in the majority of samples scores $\text{Maj@}N = 0$ even if some minority of samples were correct. Reported numbers are means of these per-problem metrics across the benchmark, optionally averaged across evaluation seeds with a sample (Bessel-corrected) standard deviation.

K Reproducibility notes

The training and evaluation code accompanying this paper is available at <https://github.com/SaFo-Lab/PW-OPSD>. The main dataset sources and evaluation conventions are as follows.

- **Datasets.** MATH-500 is HuggingFace HuggingFaceH4/MATH-500 [Hendrycks et al., 2021, Lightman et al., 2023, HuggingFaceH4, 2024]; AIME 2024 is HuggingFace HuggingFaceH4/aime_2024 [HuggingFaceH4, 2025]; AIME 2025 is HuggingFace yentinglin/aime_2025 [yentinglin, 2025]; HMMT February 2025 is a locally cleaned parquet derived from the MathArena hmtt_feb_2025 release [Dekoninck et al., 2026], SHA-256 87bfb23d2c887fab12b42fcc2b2dd8cb5a9d1070e591490dac8755bc366ea25e; the training corpus is OpenThoughts-Math-30k [Guha et al., 2025, siyanzhao, 2026], the same corpus used by Zhao et al. [2026].
- **Maj@N evaluation.** Maj@N clusters all N predictions by math-equivalence using sympy [Meurer et al., 2017] and math_verify [Kydliček, 2025], with unformatted predictions placed in a single INVALID cluster that participates in the plurality count and is scored incorrect when selected (see Appendix J for the formal definition).

L Reduction $\times$ positioning ablation

PW-OPSD differs from OPSD along two axes that are easy to conflate. The first is *positioning*: PW-OPSD multiplies the per-token loss by a position-dependent scalar $w_t \in [w_{\min}, 1]$, while OPSD uses $w_t \equiv 1$. The second is *reduction*: PW-OPSD averages the per-token loss within each rollout before averaging across the batch (a per-sequence reduction), while OPSD’s training script accumulates the loss in a global pool that effectively pools tokens across the batch first. Both changes alter how single-rollout positions contribute to the gradient, and a naive comparison conflates them. We run the full 2×2 factorial under identical training and evaluation regimes; Table 6 reports Avg@12 on AIME 2024.

Table 6: 2×2 ablation on Qwen3-4B AIME 2024: reduction (uniform vs. per-sequence) by positioning (none vs. position-weighted). Avg@12 mean $\pm$ sample standard deviation across three evaluation seeds. Bold marks the column maximum. The diagonal (OPSD and PW-OPSD Moderate) reports the same evaluation runs as the corresponding rows of Table 2; small differences in the standard-deviation digits reflect the use of sample (Bessel-corrected) standard deviation here. The position-weighted rows fix the schedule to Moderate ($w_{\min}, \tau, s) = (0.25, 0.30, 0.10)$.

Configuration	AIME 2024
Uniform / global (OPSD)	75.19 $\pm$ 0.42
Uniform / per-seq	74.72 $\pm$ 0.96
Pos-w / global	74.54 $\pm$ 0.64
Pos-w / per-seq (PW-OPSD, ours)	76.20 $\pm$ 0.58

Only the joint configuration (row 4) matches the AIME 2024 lead of Table 2; switching either axis alone underperforms by ~ 1.5 pp. The two axes are complementary on AIME 2024 rather than independently sufficient.